Hussein Sharadga

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Education

Texas A&M University, College Station, Texas| 2019 – 2022
Ph.D. in Mechanical Engineering

Jordan University of Science & Technology, Irbid, Jordan|2015 – 2017
M.Sc. in Mechanical Engineering

Al-Balqa' Applied University, Irbid, Jordan|2011 – 2015
B.Sc. in Mechanical Engineering

Academic Positions

- Current

Assistant Professor|2026 – Present
Prairie View A&M University, Mechanical Engineering

Research Affiliate|2024 – Present
University of Texas at Austin

- Past

Assistant Professor|2024 – 2026
School of Engineering, Texas A&M University- International

Postdoc Researcher|2022 – 2024
University of Texas at Austin

Research Focus

AI, GPU-Accelerated Optimization, and High-Performance Computing for Large-Scale Power Grids

Awards

ARPA-E Grid Optimization Competition- Challenge 3 Prize Winner| 2023
University of Texas at Austin
Trial 1: Ranked 2nd in the total score
Trial 2: **Ranked 1st**
Trial 3: Ranked 2nd- **$115,000 prize**

Publications

[Google Scholar](): **h-index 9; 833 citations** (Sep. 2026)

1) **H. Sharadga**, J. Mohammadi, "A GPU-based Solver for Scalable Unit Commitment in Large Power Grids", *IEEE Transactions on Industry Applications*, 2026.

2) **H. Sharadga**, Y. Du, J. Mohammadi, "Rapid Concurrent GPU-CPU Solvers for Scalable Unit Commitment in Large Power Grids", *IEEE SmartGridComm*, 2026.

3) **H. Sharadga**, Y. Du, J. Mohammadi, "Probabilistic-Adversarial Networks (PANs): A Probability-Informed Deep Learning Approach for Electrical Demand Forecasting", *IEEE Journal of Power and Energy*, 2026.

4) **H. Sharadga**, J. Mohammadi, "GPU-Accelerated Optimization Solver for Unit Commitment in Large-Scale Power Grids", *IEEE Texas Power and Energy Conference*, 2026

5) **H. Sharadga**, Abdullah Hayajneh, Erchin Serpedin, "Evaluating AI-based Image Inpainting Techniques for Facial Components Restoration Using Semantic Masks", *AI Journal*, 2026.

6) A. Dawahdeh, **H. Sharadga**, G. Zakeri, A Hayajneh, G Pritchard, "Real-time building energy storage scheduling under electrical load uncertainty: A Markov decision process approach with comprehensive analysis of different pricing policies", *Energy Reports*, 2026.

7) **H. Sharadga**, J. Mohammadi, "Scalable Unit Commitment for Large Power Grids: Relaxation, Tightening, and GPU-Based Solvers Evaluation", *IEEE North American Power Symposium*, 2025.

8) M. Golan, **H. Sharadga**, J. Mohammadi, "Power grid resilience: insights from sensitivity analysis using a comprehensive AC optimal power flow model", *Environment Systems and Decisions Journal*, 2025.

9) **H. Sharadga**, J. Mohammadi, Constance Crozier, Kyri Baker, "Scalable Solutions for Security-Constrained Optimal Power Flow with Multiple Time Steps", *IEEE Transactions on Industry Applications*, 2025.

10) **H. Sharadga**, J. Mohammadi, Constance Crozier, Kyri Baker, "Optimizing Multi-Timestep Security-Constrained Optimal Power Flow for Large Power Grids", *IEEE Texas Power and Energy Conference*, 2024.

11) **H. Sharadga**, A. Dawahdeh, "Novel MPPT Controller Augmented with Neural Network for Use with Photovoltaic Systems Experiencing Rapid Solar Radiation Changes", *Sustainability Journal*, 2024.

12) S. Hajimirza, **H. Sharadga**, "Learning Thermal Radiative Properties of Porous Media from Engineered Geometric Features", *Int. J. Heat Mass Transf*, 2021.

13) **H. Sharadga**, S. Hajimirza, Robert S Balog, "Time Series Forecasting of Solar Power Generation for Large-Scale Photovoltaic Plants", *Renewable Energy*, 2020.

14) A. Shafi, **H. Sharadga**, S. Hajimirza, "Design of Optimal Power Point Tracking Controller Using Forecasted Photovoltaic Power and Demand", *IEEE Transactions on Sustainable Energy*, 2019.

15) **H. Sharadga**, A. Dawahdeh, MA Al-Nimr, "A hybrid PV/T and Kalina cycle for power generation", *International Journal of Energy Research*, 2018.

16) M. Al-Nimr, S. Kiwan, **H. Sharadga**, "A hybrid TEG/wind system using concentrated solar energy and chimney effect", *International Journal of Energy Research*, 2018.

17) M. Al-Nimr, S. Kiwan, **H. Sharadga**, "Simulation of a novel hybrid solar photovoltaic/wind system to maintain the cell surface temperature and to generate electricity", *International Journal of Energy Research*, 2018.

Teaching Experience

Prairie View A&M University | 2026 – Present
Assistant Professor

- Dynamics, Computer-Aided Design, and Measurement & Instrumentation Lab.

Texas A&M International University | 2024 – 2026
Assistant Professor

- AI, Optimization, Smart Power Grid, Statistics, Eng. Modeling & Design, and CAD.
- **Student Evaluations: 4.75/5** (Average)

University of Texas Permian Basin | 2023 – 2024
Visiting Assistant Professor

- Thermodynamics, Fluid Mechanics, Heat Transfer, Engineering Graphics, Computer-Aided Mechanical Design, Dynamics, and Thermo-fluid & Mech Sys Lab.
- **Student Evaluations: 4.23/5** (Average) **[Evaluation Report]**

Mentored Students

Karol Gonzalez, B.Sc. Student, 2026, Texas A&M International University.
Julio Martinez, B.Sc. Student, 2026, Texas A&M International University.
Maureen S. Golan, Ph.D. Student, 2023, The University of Texas at Austin.

Professional Activities & Service

1. Invited Talks

ARPA-E Grid Software Annual Review, Electric-Stampede's Approach: Fast and Robust Strategies for Large-scale mixed-integer SCOPF.

2. Poster Presentations

Electric Stampede: A Robust Solution to The Challenging Task of Us Power Grid Optimization, INFORMS Annual Meeting, Phoenix (2023).

Time Series Forecasting of Solar Power Generation for Large-Scale Photovoltaic Plants, Texas A&M Conference on Energy (2019).

A Fast and Accurate Single Diode Model for Photovoltaic Design, IMECE, Utah (2019).

3. University Service

Search Committee Member | 2024
Texas A&M International University

ABET Committee Member | 2024
Texas A&M International University

ABET Committee Member | 2023
University of Texas Permian Basin

4. Conference Service

Session Chair & Reviewer | 2026
IEEE Texas Power and Energy Conference (TPEC 2026)

Session Chair | 2025
IEEEE North American Power Symposium (NAPS 2025)

Research Funding Activity

Probability-Informed Deep Learning for Data Center Forecasting with Probabilistic-Adversarial Networks
PI: **Hussein Sharadga**, Budget: $200,000
Submitted to: NSF

Industry Experience

Handshake AI | AI Trainer – Computer Science Expert

Remote | Oct. 2025 – Sep. 2026

Selected for specialized AI training programs for **OpenAI** and other AI labs, focused on improving large language models through expert evaluation and feedback.

Programming Skills

Languages: Python, MATLAB, Julia, C/C++.

AI/ML: Keras, PyTorch, Scikit-learn.

Optimization: Gurobi, MOSEK, IPOPT, COPT, HiGHS.

HPC: GPU-accelerated computing for large-scale optimization; parallel computing.

References

1. Dr. Javad Mohammadi
   Assistant Professor
   The University of Texas at Austin
   Email: javadm@utexas.edu
   Website: Link

2. Dr. Juan Carlos Baltazar
   Professor
   Texas A&M University
   Email: jbaltazar@arch.tamu.edu
   Website: Link

3. Dr. Kyri Baker
   Associate Professor
   University of Colorado Boulder
   Email: Kyri@colorado.edu
   Website: Link

4. Dr. Constance Crozier
   Assistant Professor
   Georgia Institute of Technology
   Email: constance@gatech.edu
   Website: Link